

ELECTRICAL SAFETY INTERVENTION REPORT #NM-2026-121

Department: Plant Facilities & Maintenance

Location: Substation Alpha - Main 480V Switchgear Room

Equipment Involved: 480V Motor Control Center Feeder Breaker (MCC-3)

Reporter Role: Master Electrician

Immediate Action Taken: Feeder tripped manually, load shifted to secondary transformer

Incident Description and Narrative:

During scheduled thermographic infrared inspection of Substation Alpha, the B-phase terminal lug on 480V feeder breaker MCC-3 showed extreme thermal discoloration and measured 198 degrees Celsius under 70% load. Terminal bolt was torqued to only 15 ft-lbs against 45 ft-lbs spec. High vibration from nearby compressor room had loosened connection. Any transient current surge during motor start would have triggered a catastrophic arc flash blast inside the enclosed electrical room.

Investigation Findings and Safeguard Assessment:

Physical inspection confirms barrier defect and procedural violation. Immediate review of lockout/tagout and secondary containment is mandated before resuming operations.